# ADITHYAN R S

AI Enthusiast, Systems Thinking & Applied Machine Learning
Trivandrum, Kerala | +91 97466 54062 | adirs5022@gmail.com
github.com/adi5022 | linkedin.com/in/adimbcet

## SUMMARY

AI enthusiast with strong systems-level understanding of modern machine learning architectures, including retrieval-augmented generation, LLM agents, and mechanistic interpretability. Focuses on system design and architecture, pairing conceptual grounding in how these systems work with AI-assisted engineering to build and iterate on complex projects efficiently, while building hands-on programming fundamentals in parallel.

## EDUCATION

**B.Tech, Computer Science and Engineering (AI)**
Mar Baselios College of Engineering & Technology, Thiruvananthapuram 2023 - 2027

**Higher Secondary (12th), CBSE**
Loyola School, Sreekariyam, Thiruvananthapuram 2023

## SKILLS

**Core Programming:** Python, C | **Learning:** C++ (basic), C# (basic)
**AI / ML Concepts:** Retrieval-Augmented Generation, LLM agents, Prompt Engineering, mechanistic interpretability (sparse autoencoders, activation steering, model editing), classical ML (regression, gradient boosting), Deep Learning, Supervised Learning, Generative AI, Transformer Architecture, Computer Vision
**Frameworks & Tools:** LangGraph, FAISS (Vector Database), NetworkX, PyTorch, TransformerLens, SAELens, Flask, Streamlit, PyQt6, Groq API
**CS Fundamentals & Tooling:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Git, REST APIs, SQLite

## PROJECTS

**FeatureScalpel: Mechanistic Interpretability and Intervention of LLMs via Sparse Autoencoders (Ongoing)** *Python, PyTorch, TransformerLens, SAELens, CUDA* github.com/adi5022/Mechanistic-Intervention...

- Built a research pipeline that edits GPT-2's residual stream along sparse autoencoder feature directions to correct factual errors at inference time, with no retraining or weight updates, and can explain the full mechanism in depth against fine-tuning, prompting, and ROME-style weight edits.
- Designed the benchmark methodology comparing this approach against activation steering and ROME across 22 factual-editing prompts from ROME's CounterFact dataset, evaluated across transformer depth (layers 0-11).
- Currently building a GPU-batching optimization to reduce the number of forward passes required per prompt-layer evaluation; performance gains and equivalence against the original sequential approach are still being validated.
- Ran the project through structured, session-logged agentic coding workflows (Claude Code), maintaining handoff documentation and validation scripts across iterations.

**MBAssist AI: Multilingual RAG Admissions Assistant** *Python, Flask, LangGraph, FAISS, Groq (Llama-3.3-70B), Sarvam AI* github.com/adi5022/MBAssist-AI

- Built a voice-enabled, multi-agent RAG chatbot for college admissions, using LangGraph-orchestrated retrieval over scraped institutional documents and a FAISS vector index.
- Designed a hybrid memory architecture, sliding-window recent context plus LLM-summarized long-term archive, to bound token usage in long conversations.
- Built multilingual speech-to-text support (English and Malayalam) via Sarvam AI and Groq Whisper, plus a per-user API-key model that keeps hosting costs at zero.
- Set up automated weekly re-indexing through a background scheduler and a REST admin endpoint, using thread-safe locks to prevent concurrent rebuilds.

**GraphRAG Research Assistant: Hybrid Vector + Graph RAG System** *Python, Flask, FAISS, NetworkX, Groq/Gemini*github.com/Tharun-1-6/graph-rag-research-assistant

- Co-built a hybrid RAG system for querying scientific literature, combining a FAISS vector index with a NetworkX knowledge graph for factual and multi-hop relational queries.
- Designed the query routing logic that classifies incoming queries and selects between vector retrieval, graph retrieval, or a combined pipeline, integrating the two independently-built retrieval systems into one.
- Built the interactive graph presentation layer, rendering the knowledge graph and per-query retrieval traces (route chosen, nodes traversed, latency) for transparency, plus smaller fixes across the wider system alongside a teammate.

**FlowState AI: Cardiovascular Digital Twin** *Python, PyQt6, scikit-learn, Plotly, Gemini/Groq* github.com/adi5022/flow-state-AI

- Designed a patient-specific cardiovascular digital twin architecture projecting blood pressure, kidney function (eGFR), and survival probability over a 30-year horizon from 10 clinical inputs, with every coefficient grounded in peer-reviewed literature, including a two-layer eGFR decline model (clinical equations plus a gradient boosting residual) achieving cross-validated $R^2$ of 0.957.
- Modeled probabilistic ACE-inhibitor intervention effects to visualize treated vs. untreated divergence, and contributed to integrating an LLM assistant (Gemini/Groq) for explaining results in plain language, as part of a 4-person team.

**Personal Portfolio Website** *HTML, CSS, JavaScript, SVG* github.com/adi5022/My-Personal-Portfolio-Website

- Built a neuron-themed portfolio site featuring a scroll-driven SVG axon-growth animation and a custom contact speed-dial widget, implemented in vanilla JavaScript.